

Insurance Premium Pricing & Risk Factor Analysis

Independent Analyst Project — Health Insurance Risk Segmentation

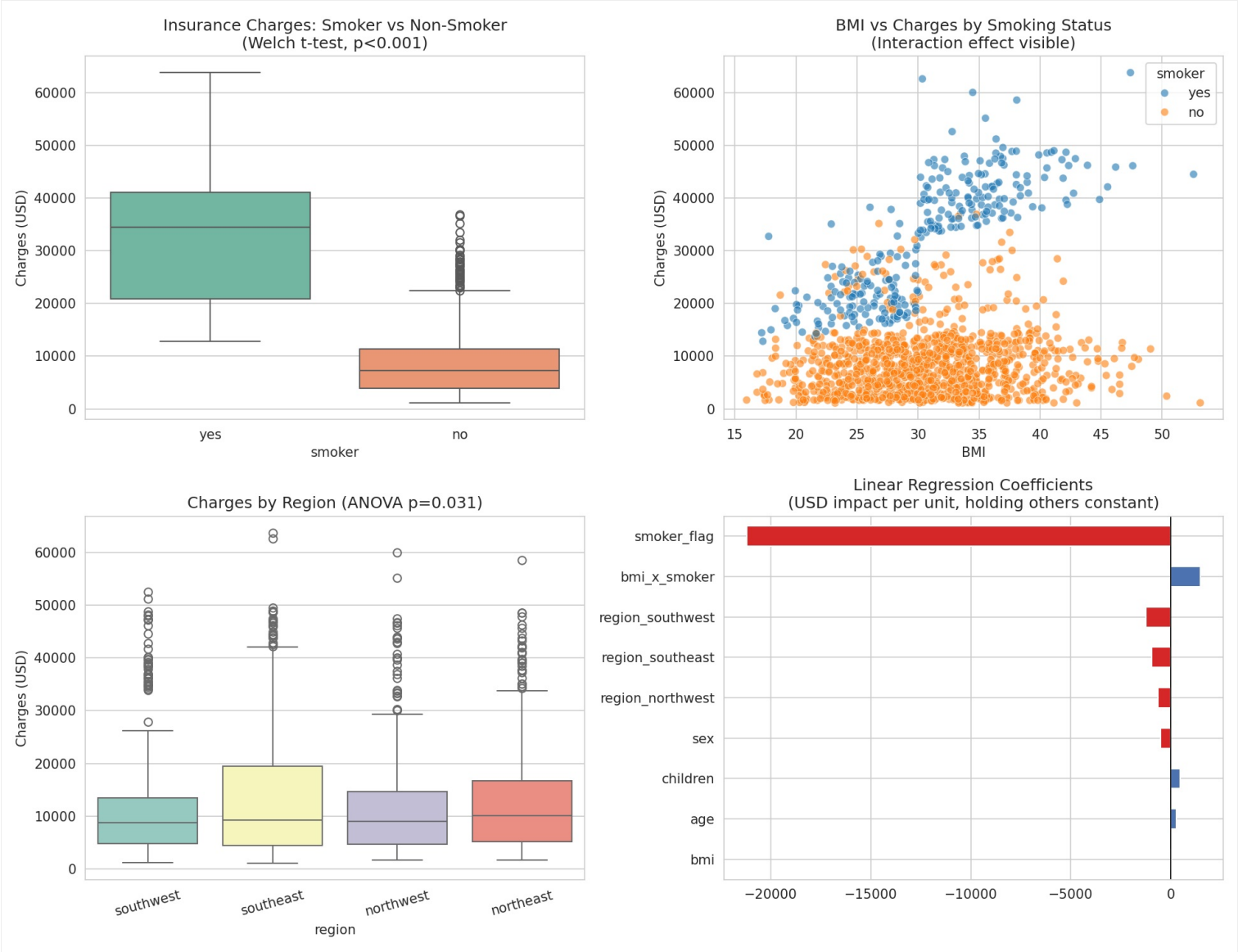
Dataset: Medical Cost Personal Datasets — 1,338 U.S. health insurance beneficiary records (age, sex, BMI, children, smoker status, region, actual billed charges). Publicly available reference dataset, originally published in *Machine Learning with R* (Lantz, Packt Publishing).

1. Objective

Insurance pricing depends on correctly identifying which risk factors drive cost, and how much each one matters. This project validates candidate risk factors using hypothesis testing, then builds an interpretable pricing model to quantify their dollar impact — the analytical approach underlying premium tiering and underwriting risk segmentation.

2. Risk Factor Validation (Hypothesis Testing)

Figure 1: Statistical Test Results



Test	Result	Interpretation
Smoker vs. non-smoker charges (Welch's t-test)	$t = 32.75, p < 0.001$	Smokers pay 3.8x more on average (\$32,050 vs. \$8,434)
BMI vs. charges (overall) (Pearson correlation)	$r = 0.198, p < 0.001$	Weak on its own — misleading in isolation
BMI vs. charges, smokers only	$r = 0.806, p < 0.001$	Very strong once smoking is present

BMI vs. charges, non-smokers only	$r = 0.084, p = 0.006$	Negligible practical effect
Regional differences (one-way ANOVA)	$F = 2.97, p = 0.031$	Statistically significant, practically small

Key Insight: BMI is not a meaningful standalone pricing factor — its effect on cost is almost entirely *conditional* on smoking status. Tested in isolation ($r=0.198$), BMI looks unimportant. Split by smoking status, a sharply different picture emerges ($r=0.806$ for smokers vs. $r=0.084$ for non-smokers). This is a classic case where an averaged statistic hides the real driver of risk.

3. Pricing Model (Interpretable Linear Regression)

A linear regression model was fit with an explicit **BMI × Smoker interaction term** to directly quantify the effect identified above, rather than relying on a black-box model.

Model performance: $R^2 = 0.865$, $MAE \approx \$2,757$

- **Being a smoker** adds a base premium of roughly **\$21,200** versus a non-smoker with otherwise identical characteristics.
- **Each additional BMI point** adds only ~\$20 to charges for a non-smoker, but adds an **extra \$1,471** on top of that for a smoker — confirming the interaction effect is not just statistically significant, but large in dollar terms.
- Age, number of children, and region contribute comparatively minor, secondary effects.

4. Interactive Dashboard (Tableau)

A companion interactive dashboard was built in Tableau Public to allow exploration of these relationships by region, BMI category, and smoking status: [https://public.tableau.com/app/profile/yongyi.ye/viz/Insurance_Risk_Analysis/Dashboard1?publish=yes]

5. Business Recommendation

Underwriting models that price BMI and smoking as independent, additive factors will systematically **overprice healthy-weight smokers** and **underprice obese smokers**. Because the two factors interact rather than add, premium structures should apply a joint **smoker × BMI-band** pricing tier rather than separate linear surcharges — this more accurately reflects each segment's true cost risk.

6. Methodology & Tools

Analysis: Python (pandas, scipy for hypothesis testing, scikit-learn for regression) | **Visualization:** matplotlib/seaborn (statistical charts), Tableau Public (interactive dashboard)

Full analysis code, dataset, and raw statistical output available in this repository (analysis_v2.py, results_summary_v2.txt).